

PRECISION AGRICULTURE

Last 5 Year Previous Year Question (2019-2023)

1. Which farming method is an information and technology-based agricultural management system to identify, analyse and manage soil-spatial and temporal variability within fields for optimum profitability, sustainability and protection of the environment? (2020,2022)
 - a. Precision farming
 - b. Organic farming
 - c. Sustainable agriculture
 - d. None of the above

Ans A.

Question in 2023 mains

1. Explain the 5 uses of technology in agriculture with the help of government initiative to adopt technology in agriculture.

PRECISION AGRICULTURE

It is the management approach to the cultivation of the crops **by considering spatially and temporal variations** in soils, rainfall, and adopting management techniques to reduce cost, efficient use of inputs and reduce environmental pollution.

Precision agriculture is also called as precision farming in which in field variability is taken into account and according to the local circumstances of a given field, seeding, nutrient management, plant protection, measures etc are taken up which improves the resource utilization in sustainability in the system.

VARIOUS DEFINITIONS OF PRECISION FARMING

Precision agriculture can be defined as the application of principles and technologies to manage spatial and temporal variability associated with all aspects of agricultural production for the purpose of improving crop performance and environment equally.

Or

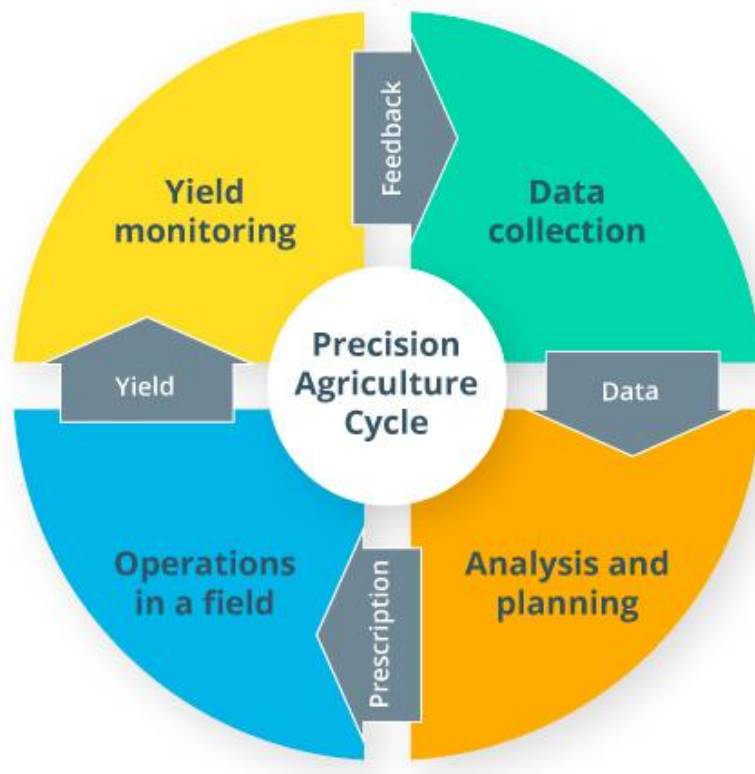
Precision farming is also **referred as a site-specific farming, smart farming** and GPS based farming.

Or

Precision farming is the only solution to identify the causes of variability within the field and to carefully tailor soil and crop management to fit in each cultivated field.

CHARACTERISTICS OF PRECISION AGRICULTURE

1. **Timely collection of the spatial & temporal** data from a field.
2. **Proper analysis** of the spatial and temporal variations in the field for crops, soils, & environment.
3. **Application of new technology** like global positioning system (GPS), geographical information system (GIS), remote sensing, yield monitors etc.
4. It is based on **information and knowledge for the scientific management** of modern agriculture.
5. All the **available resources are fully utilised** to promote sustainable agriculture.



Flow chart 1- important aspects of precision agriculture

PRINCIPLES OF PRECISION FARMING

1. **First principle is to understand the basic principle of variability in both time and space**- as there is variation found in all agricultural fields and may vary in soil fertility, moisture content, soil texture, topography, pest population, pathogen intensity, weed density etc.
 - ✓ **Variability in time**- like nutrient recommendations for crop may not be required for next season crop.
 - ✓ **Variability in space**- more weed infestation in the middle of the fields as compared to the border areas.
2. **Second principle is the timely collection and analysis of the spatial and temporal variant information** of crops, soil, and environment by using the relevant information technologies like remote sensing.
 - ✓ Proper analysis of the data using appropriate technologies for decision making.

3. Third principle underlies with precision agriculture is managing the variability with the site-specific agronomic recommendations.

After thoroughly analysis of the field the **field is to be divided into proper management zones** such as

- ✓ High yield management zones,
- ✓ Low yield zone,
- ✓ High fertility zones,
- ✓ Acidic zones etc.
- ✓ High weed and pest incidence.

Decision is to be **taken on the basis of the local circumstances** of a given field like seeding, nutrient managements, plant protection measures etc.

TECHNOLOGIES USED IN PRECISION AGRICULTURE

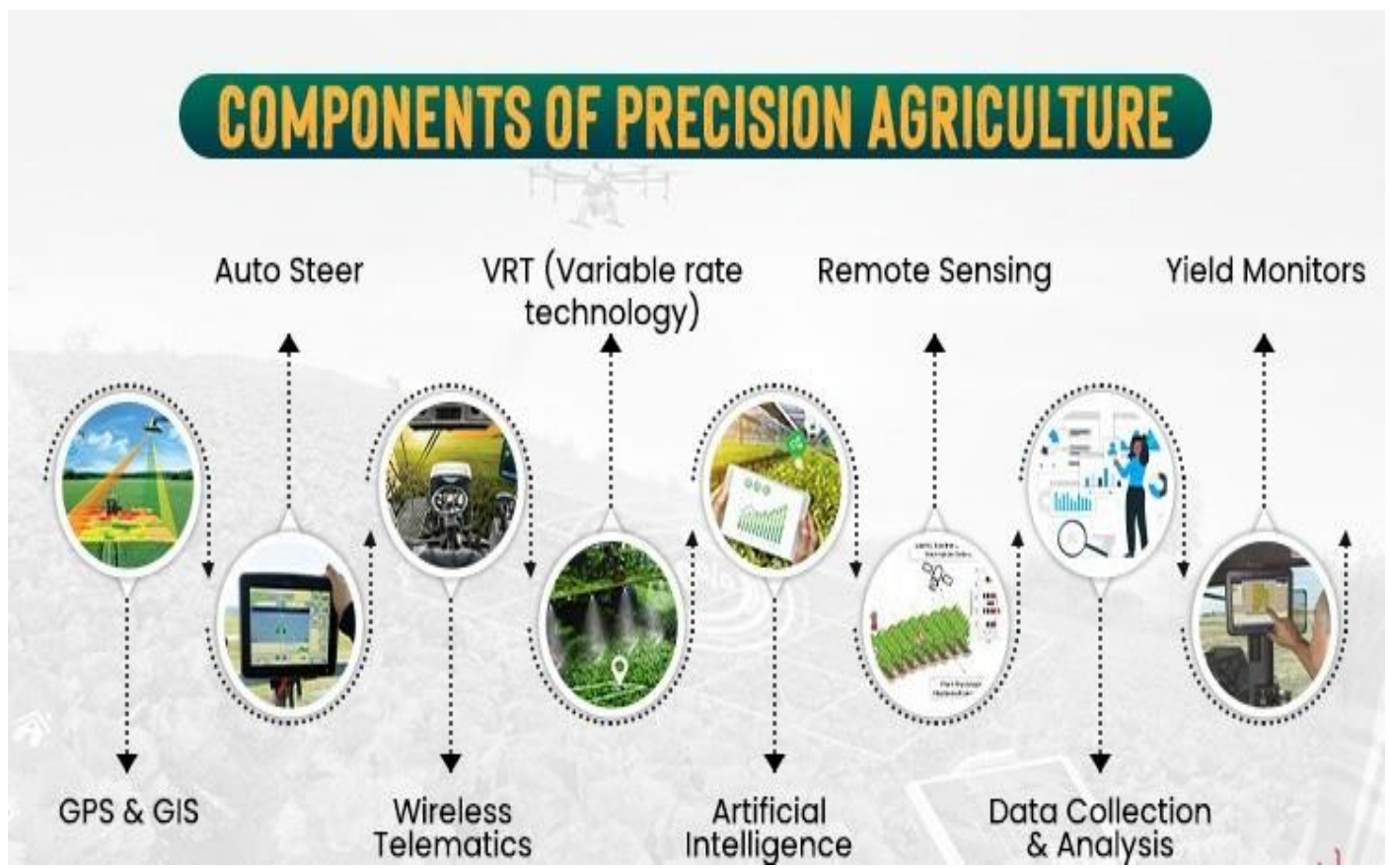


Image (i)- technologies used in precision agriculture

1. GLOBAL POSITIONING SYSTEM (GPS)-

The Global Positioning System (GPS) is a **space-based satellite navigation** system that provides location and **time information in all weather conditions, anywhere on or near the Earth** where there is an unobstructed line of sight to four or more GPS satellites. GPS works in any weather conditions, anywhere in the world, 24 hours a day.

Application Of GPS In Agriculture

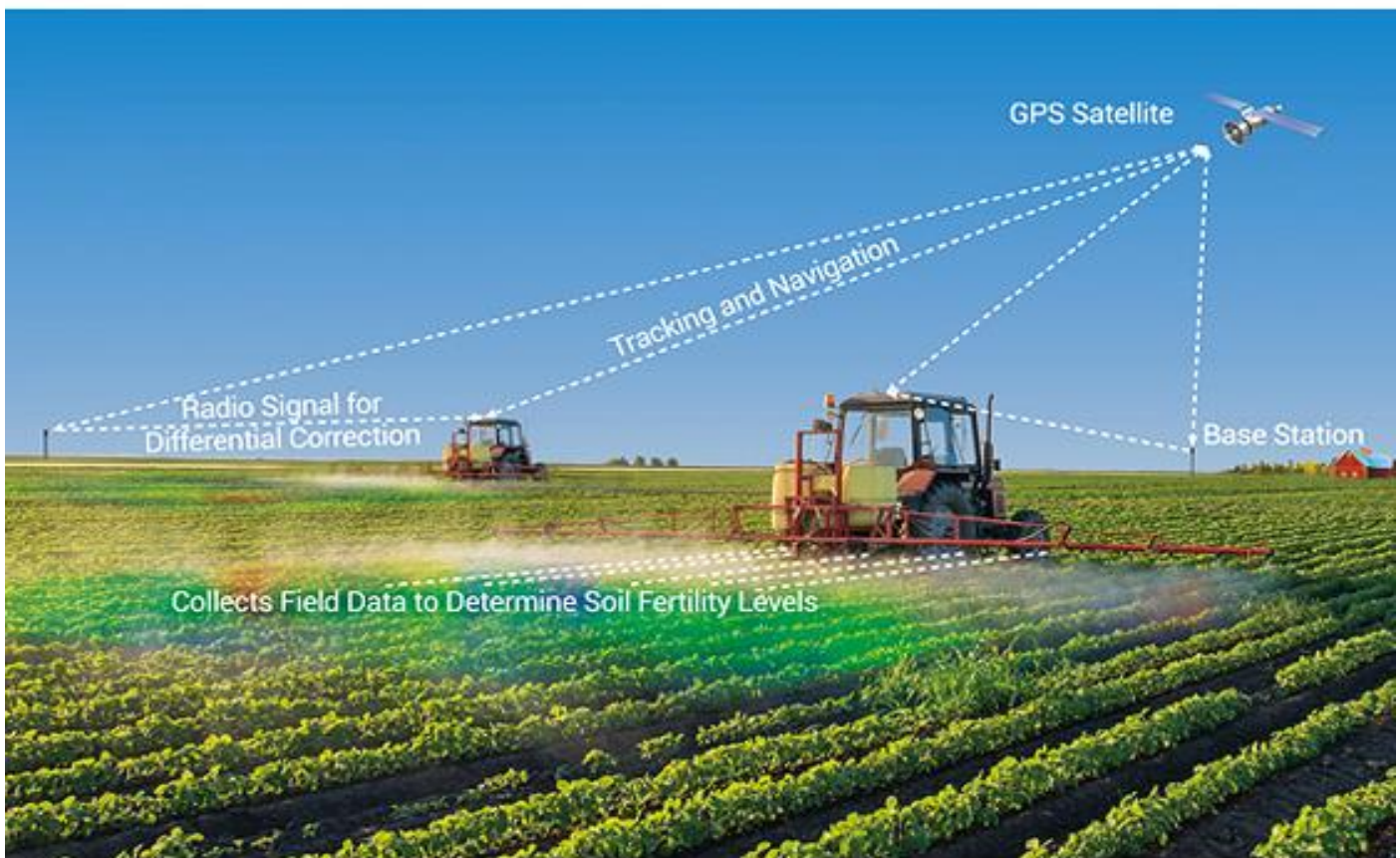


Image (iii)- use of GPS in agriculture

- a. GPS-server.net service can be **used for precision farming, field planning, yield mapping, and tractor guidance.**
- b. The accuracy of GPS allows farmers **to create farm maps with precise acreage for field areas**, road locations and distances between points of interest.

- c. Crop dusters equipped with GPS devices are able to spread swaths accurately over the field, **applying chemicals only where needed**, minimizing chemical drift, reducing the number of chemicals needed, thereby benefiting the environment and more accurate usage of materials.
- d. GPS allows farmers to **accurately navigate to specific locations** in the field, year after year, to collect soil samples or monitor crop conditions.
- e. Location information is collected by GPS receivers for mapping field boundaries, roads, irrigation systems, and problem areas of crops such as weeds or disease.

2. REMOTE SENSING-

Remote sensing is the **process of acquiring information about the Earth's surface by measuring its reflected and emitted radiation** without coming into direct contact with the object. In much of remote sensing, the process involves an interaction between incident radiation and the targets of interest

Remote sensors can be either passive or active.

- a. **Passive sensors** respond to external stimuli. **They record natural energy that is reflected or emitted from the Earth's surface.**

The most common source of radiation detected by passive sensors is reflected sunlight.

- b. **Active sensors** use internal stimuli to collect data about Earth.

For example, a **laser-beam remote sensing system** projects a laser onto the surface of Earth and measures the time that it takes for the laser to reflect back to its sensor.

Application Of Remote Sensor in Agriculture

- a. Remote sensing is used to **forecast the expected crop production** and yield over a given area and determine how much of the crop will be harvested.
- b. In the event of **crop damage or crop progress**, remote sensing technology can be used to penetrate the farmland and determine exactly how much of a given crop has been damaged.

- c. Remote sensing also **allows farmers and experts to predict the expected crop yield** from a given farmland by estimating the quality of the crop and the extent of the farmland.
- d. Role in **identification of pests in farmland** and gives data on the right pests control mechanism to get rid of the pests and diseases on the farm.
- e. Remote sensing **gives the soil moisture data and helps in determining the quantity of moisture** in the soil.
- f. **Soil mapping is one of the most common yet most important uses of remote sensing.** Through soil mapping, farmers are able to tell which soils are ideal for which crops and which soil require irrigation and which ones do not. This information helps in precision agriculture.

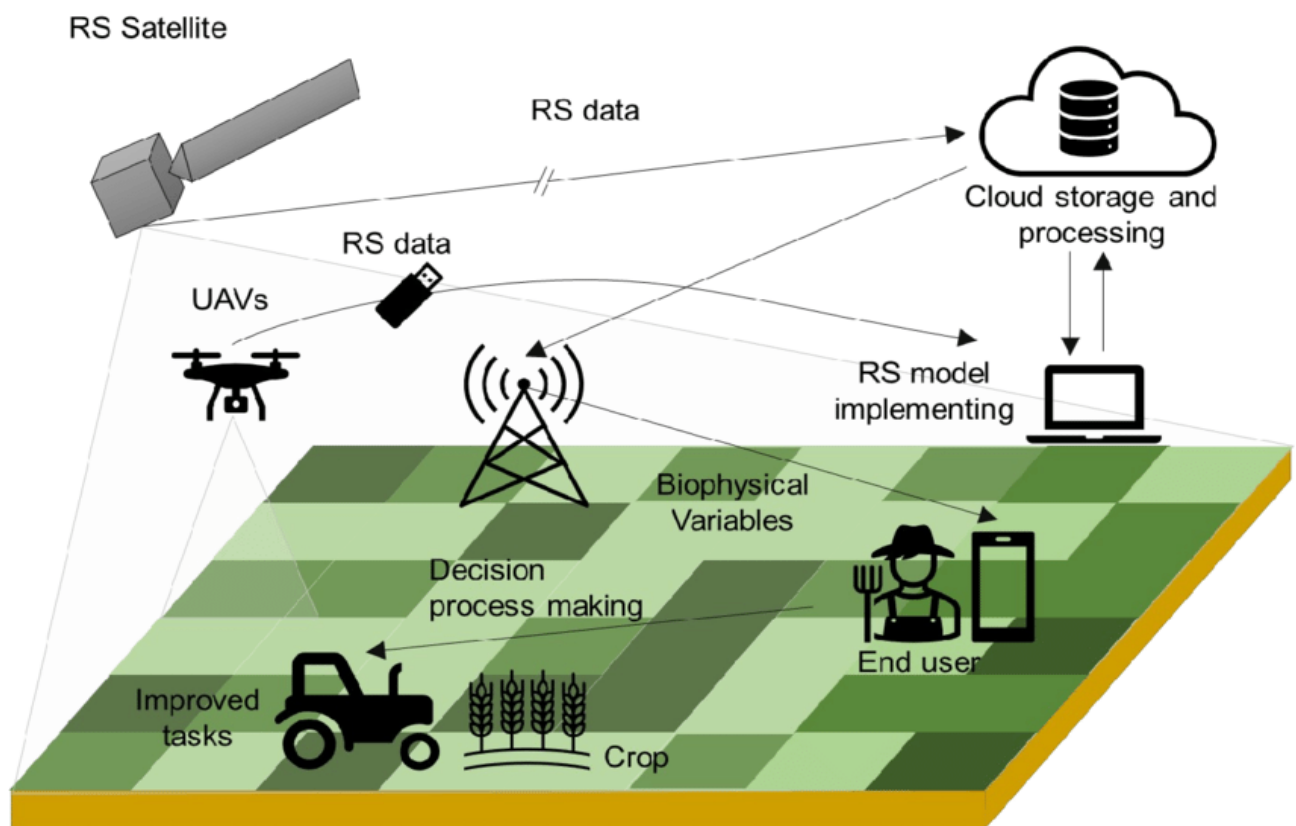


Image (iii)- use of remote sensing in agriculture

3. GEOGRAPHICAL INFORMATION SYSTEM (GIS)

A geographic information system (GIS) is a **computer system for capturing, storing, checking, and displaying data related to positions on Earth's surface**. By relating seemingly unrelated data, GIS can help individuals and organizations **better understand spatial patterns and relationships**. GIS can use any information that includes location.

Application Of GIS In Agriculture

- a. **Can identify vegetation level in the field** and this information can be used for application of fertilizer, irrigation, weedicide etc.
- b. Can be **used to prepare soil and field maps**.
- c. This technology can be **used to check the crop health, pest attack, weed** incidence in the fields etc.

4. DRONES

A drone is an unmanned aircraft. **Drones are more formally known as unmanned aerial vehicles (UAV)** or unmanned aircraft systems. Essentially, **a drone is a flying robot that can be remotely controlled or fly autonomously** using software-controlled flight plans in its embedded systems, that work in conjunction with onboard sensors and a global positioning system (GPS).

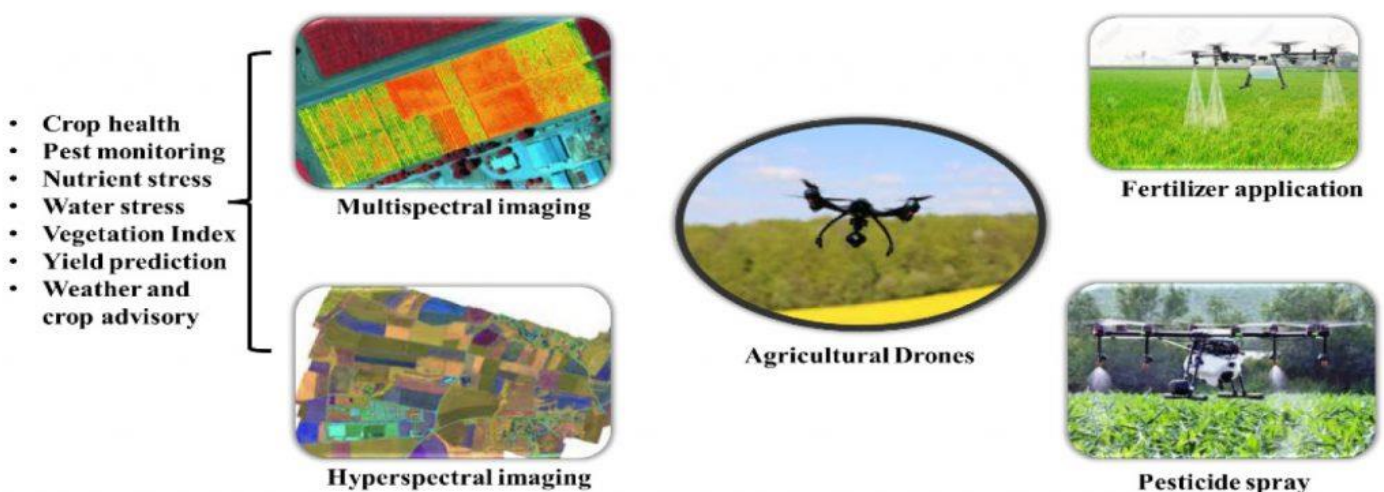


Image (iv)-use of drone in agriculture

Application of drones in agriculture

- a. For **efficient field planning**, agricultural drones can be used for soil and field analysis.
- b. They can be used to mount sensors **to evaluate moisture content** in the soil, terrain conditions, soil conditions, soil erosion, nutrients content, and fertility of the soil.
- c. **Crop surveillance** is the **supervision of crop progress from the time seeds are sown to the time for harvest**. This includes providing fertilizers at the right time, checking for pest attack, and monitoring the effect of weather conditions.
- d. Drones **can help in planting trees and crops**, which was done by farmers before. This technology will not only save labour but also help in saving fuels.
- e. Agri-drones **can be used to spray chemicals as they have reservoirs**, which can be filled with fertilizers and pesticides for spraying on crops in very little time, as compared to traditional methods.

ADVANTAGES OF PRECISION AGRICULTURE

- 1. **Reduction In Cost of Cultivation**- by applying production inputs based on site specific needs of the fields, the excessive and indiscriminate use of chemicals/inputs can be avoided. Hence there is less wastage of the inputs.
- 2. **Increases Input Efficiency**- timely application and in proper quantity not only reduce wastage but an increased input efficiency is visible on the field by providing more yield.
- 3. **Reduction In Population**- site specific application of the inputs depending upon the spatial and temporal variations, indiscriminate and excessive usage of all production input including water there by safeguard the environment from pollution.

4. **Reduction in application of nutrients**- especially nitrogen fertilizer thus reducing nitrate in underground water and nitrous oxide to the atmosphere.
5. **Reduction in chemical doses** through variable rate application technology.
6. **Reduction in the application** of the irrigation water, thus reducing leaching of nutrients along with deep percolation.
7. **Reduce erosion, runoff, and sedimentation of the water bodies.**

DISADVANTAGES OF PRECISION AGRICULTURE

1. **New and emerging technology.**
2. **Require high amount of investment** which is not much feasible in Indian agriculture as most of the farmers have 2 ha average land holding.
3. **Positive benefit will only be visible after long term** usage of technology in the field.
4. **Lack of technical know-how** among the farmers.
5. **Loop holes in implementation** of government policies at the ground level.
6. **Less research and development** in this sector.

ROLE OF TECHNOLOGY IN AGRICULTURE

Agriculture in India is largely dependent on nature, but climate and global warming issues make farming unpredictable. **The need of the hour is to educate farmers in the use of modern technology** and innovative approaches to increase productivity and raise profitability.

1. Improved productivity from the mechanization of agriculture-

Agricultural mechanization has the potential to, directly and indirectly, affect yields through a reduction in post-harvest losses and an increase in harvest gains.

2. Climate/ weather prediction through artificial intelligence-A major advance in agriculture is the use of artificial intelligence (AI).

- ✓ **Modern equipment and tools based on AI enable data gathering and assist in precision farming and informed decision-making.**
- ✓ Drones, remote sensors, and satellites gather 24/7 data on weather patterns in and around the fields, providing farmers with vital information on temperature, rainfall, soil, humidity, etc.

3. Resilient crops developed via the use of biotechnology-Agriculture refers to a wide resource of methodologies that include traditional breeding methods, genetic engineering, and the development of microorganisms for agriculture.

- ✓ **Genetic engineering uses the understanding of DNA to identify and work with genes to increase crop resistance to pests, and the development of high-yielding varieties also makes improvements to livestock.**

4. Agriculture Sensors – Communications technology has evolved rapidly in India and made smart farming a possibility.

- ✓ **Sensors are now being used in agriculture to provide data to farmers to monitor and optimize crops given the environmental conditions and challenges.**
- ✓ These sensors are based on wireless connectivity and find application in many areas such as determining soil composition and moisture content, nutrient detection, location for precision, airflow, etc.
- ✓ **Sensors help farmers save** on pesticides, and labour, and result in efficient fertilizer application. They allow farmers to maximize yields using minimal natural resources.

5. Improving farm yields and supply chain management use Big Data-

The collection and compilation of data and its further processing to make it useful for decision-making/problem-solving are expanding the way big data functions.

- ✓ Big data is slated to play a major role in smart farming, and the benefits percolate across the entire supply chain and the markets. Agriculture is becoming larger, and it depends on a large number of variables.

6. Livestock monitoring – The use of chips and body sensors can help prevent disease outbreaks and are crucial in large-scale livestock management.

- ✓ Chips and body sensors measure vital parameters and indicators that could detect illness early and prevent herd infection.
- ✓ Ultrasounds are a useful tool to judge the quality of meat. This helps control and improve the quality of the meat.

7. Monitor and Control Crop Irrigation Systems through Smartphones-

mobile technology has also been playing a significant role in monitoring and controlling crop irrigation systems.

- ✓ With this modern technology, farmers can control their irrigation systems via smartphones and computers instead of driving to each field.

- ✓ Moisture sensors planted underground can provide information regarding the moisture levels present at certain depths in the soil.

GOVERNMENT INITIATIVE FOR DIGITAL AGRICULTURE IN INDIA

Government has taken various initiatives to give a push to digital agriculture in the country, which are given below:

1. Government has finalized the core concept of **India Digital Ecosystem of Agriculture (IDEA) framework** which would lay down the architecture for the federated farmers' database.
2. **National e-Governance Plan in Agriculture (NeGP-A)** wherein, funds are released to the State(s)/UT(s) for project involving use of modern technologies viz. Artificial Intelligence (AI), Machine Learning (ML), Robotics, Drones, Data Analytics, Block Chain etc.
3. **Sub Mission on Agricultural Mechanization (SMAM)**- The scheme aims at 'reaching the unreached' by bringing to the small and marginal farmers in the core and giving the benefits of farm mechanization,
 - ✓ **Promoting 'Custom Hiring Centres'**, creating hubs for hi-tech & high value farm equipment's, distribution of various agricultural equipment's,
 - ✓ **Creating awareness** among stakeholders through demonstration and capacity building activities.
 - ✓ **Ensuring performance**- testing and certification at designated testing centres located all over the country.
4. **National Agriculture Market (e-NAM)** is a pan-India electronic trading portal which networks the existing Agricultural Produce Market Committee (APMC) mandis to create a unified national market for agricultural commodities.
 - ✓ Digital services are provided to traders, farmers, Farmers Producer Organizations (FPO), Mandis through various modules of e-NAM platform such as FPO trading module, warehouse-based trading module.

5. **Integrated Scheme for Agricultural Marketing schemes (AGMARKNET)**

to promote creation of agricultural marketing infrastructure by providing backend subsidy support to State, cooperative and private sector investments Services are provided **through (AGMARKNET) portal which is a G2C e-governance portal** that caters to the needs of various stakeholders such as farmers, industry, policy makers and academic institutions by providing agricultural marketing related information from a single window.

- ✓ It facilitates web- based information flow, of the daily arrivals and prices of commodities in the agricultural produce markets spread across the country.

6. **Agriculture Infrastructure Fund (AIF):** To mobilize a medium - long term debt finances facility for investment in viable projects for post-harvest management Infrastructure and community farming assets through incentives and financial support in order to improve agriculture infrastructure in the country. **Financial assistance is provided digitally in the form of Interest Subvention and Credit Guarantee for setting up post-harvest management Infrastructure** to beneficiaries such as Farmers, Primary Agricultural Credit Societies (PACS), Farmer Producers Organisations (FPOs), Self Help Groups (SHG), State Agencies/APMCs.

7. **Development of Kisan Suvidha mobile application** to facilitate dissemination of information to farmers on the critical parameters viz., Weather; Market Prices; Plant Protection; input Dealers (Seed, Pesticide, Fertilizer) Farm Machinery; Soil Health Card; Cold Storages & Godowns, Veterinary Centres and Diagnostic Labs.

- ✓ With market information, farmers are better informed about markets to sell produce, prevailing market prices and quantity demanded in the market. Thus, they can make informed decisions to sell produce at the right price and right time.

8. **Government is providing advisories services** on various crop related matter to the registered farmers through SMSs.

9. **India Meteorological Department (IMD) runs an operational Agrometeorological Advisory Services (AAS) viz., Gramin Krishi**

Mausam Sewa (GKMS) scheme for the benefit of farming community in the country.

- ✓ Under the scheme, medium range weather forecasts at district and block level are generated and based on these forecasts, 130 Agromet Field Units (AMFUs), located at State Agricultural Universities, institutes of Indian Council of Agricultural Research (ICAR) and Indian Institute of Technology (IIT) etc.

10. India Digital Ecosystem of Agriculture (IDEA) which would lay down the architecture of **Digital Agri Stack with unique ID for farmers**, and standards for interoperability between various data streams in the Agri stack

11. Krishi Vigyan Kendras (KVKs) and Agricultural Technological Management Agency (ATMA) at district level undertake training, demonstrations, exhibitions and skill development programs etc. provide information to the farmers, farm women, and rural youth.

12. Government is actively encouraging and promoting various innovations in the agriculture sector through different schemes/programs like

- ✓ Digital agriculture mission
- ✓ Kisan drones
- ✓ Enam
- ✓ Agri-startups
- ✓ Mechanization
- ✓ Solar pumps
- ✓ Micro-irrigation
- ✓ Bio-fortification.
- ✓ The input efficient technology like use of laser guided land leveller for increasing water use efficiency,
- ✓ Roto-till-drill and zero till drill for sowing wheat under no tillage conditions
- ✓ Broad- bed-furrow seeder for farming broad bed and seeding on bed
- ✓ Ridge-furrow seeder for sowing soybean on ridges

All these technologies and initiatives of government have increased farmers efficiency and income.

